ALPHANUMERIC SERIES AND NUMBER PUZZLES

Letters, Numbers, Symbols and Digit Positions

OAVS TGT | Reasoning (Additional topic from the “etc.” in the syllabus) | Topic 25

Syllabus line (Reasoning, 30 Marks): The verbal and non-verbal topics listed in the syllabus end with “etc.”. This topic is not named in that line, but it is asked in the papers: letter-number-symbol series (Mathematics 2024 Q83) and three-digit number puzzles (Science 2024 Q81, Q94) appear.

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). Three questions appear (Maths 2024 Q83, Science 2024 Q81 and Q94). All three are solved below. The practice questions were generated or checked by program so that each has exactly one correct option.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	The 3 PYQs and their conditions
2	Foundation	Reading a series, digit positions, highest/lowest
3	Solved PYQs	3 PYQs with step-by-step reasoning
4	Shortcuts, Traps and Scope	Speed methods and mistakes
5	Practice Set	6 original questions with a verified key
6	Quick Revision	One-page cheat sheet

CHAPTER 1

PYQ Analysis

Three questions, two papers

Paper	Q.	Question	Type	Ans
Mathematics 2024	83	How many symbols are immediately preceded by a letter and also immediately followed by a number in A # 1 * + B – D 6 7 K % @ & Y 3?	Counting in a series	D (One)
Science 2024	81	Third digit of the lowest number subtracted from the third digit of the highest in 511 484 496 608 600	Digit position	4
Science 2024	94	Third digit of the lowest number subtracted from the second digit of the highest in 497 449 483 417 426	Digit position	3 (2)
English 2019 / English 2022 / Social Studies 2023	–	none		

PYQ INSIGHT: All three PYQs give a **list to read from left to right** and a short condition. Two are **three-digit-number puzzles** (find the highest and lowest number first, then use the digit position asked) and one is a **symbol-position count**. No difficult arithmetic: accuracy of reading is the skill.

Pattern	PYQs	Priority
Three-digit numbers: digits of the highest and lowest	2	★★★ Must
Series of letters, numbers and symbols: count with “preceded by / followed by”	1	★★★ Must
Rearranging digits or numbers by a rule	0	★★ Expected

CHAPTER 2

Foundation

Reading rules and a fixed method

2.1 Series of letters, numbers and symbols

Wording	Meaning
“immediately preceded by”	The item directly to the left
“immediately followed by”	The item directly to the right
“between two …”	Left and right neighbours both satisfy the condition
“Left” and “right”	Given with the series (here the left end is the start)

NOTE: The first and last items cannot have a neighbour on one side, so they cannot satisfy a “preceded and followed” condition.

2.2 Three-digit number puzzles

Step	Do this
1	Read the example (697: first digit 6, second 9, third 7).
2	Find the **highest** and **lowest** numbers by comparing whole numbers (hundreds first).
3	Pick the digit asked from each number.
4	Do the operations in the order stated (“A subtracted from B” means B − A).

SHORTCUT: Science 2024 Q81: highest 608, lowest 484: third digits 8 and 4, so 8 − 4 = 4. Science 2024 Q94: highest 497, lowest 417: second digit of 497 is 9, third digit of 417 is 7, so 9 − 7 = 2.

CHAPTER 3

Solved PYQs

All three questions with step-by-step reasoning

Mathematics 2024 Q83

(left) A # 1 * + B – D 6 7 K % @ & Y 3 (right). How many such symbols are there which are immediately preceded by a letter and also immediately followed by a number? (A) None (B) Three (C) Two (D) One

1. Symbols in the series: #, *, +, –, %, @, &.
2. Check each: # is preceded by A (a letter) and followed by 1 (a number): yes. * is preceded by 1 (a number): no. + is preceded by * (a symbol): no. – is preceded by B (a letter) but followed by D (a letter): no. % is preceded by K (a letter) but followed by @ (a symbol): no. @ and &: preceded by symbols: no.
3. Only # satisfies both conditions: count = 1.
Answer: Option D (One)

Science 2024 Q81

(Left) 511 484 496 608 600 (Right). What will be the result if the third digit of the lowest number is subtracted from the third digit of the highest number? (1) 7 (2) 6 (3) 3 (4) 4

1. Highest number: 608. Lowest number: 484.
2. Third digit of 608 = 8. Third digit of 484 = 4.
3. Subtract: 8 − 4 = 4.
Answer: Option 4 (4)

Science 2024 Q94

(Left) 497 449 483 417 426 (Right). What will be the result if the third digit of the lowest number is subtracted from the second digit of the highest number? (1) 5 (2) 4 (3) 2 (4) 6

1. Highest number: 497. Lowest number: 417.
2. Second digit of 497 = 9. Third digit of 417 = 7.
3. Subtract: 9 − 7 = 2.
Answer: Option 3 (2)

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and mistakes

- **Mark the neighbours:** write the condition (letter, symbol, number) on the left and right of each candidate.
- **Compare whole numbers:** the highest has the largest hundreds digit; if tied, compare tens.
- **Name digits by position:** first = hundreds, second = tens, third = units.
- **Order of subtraction:** “A subtracted from B” means B − A.
- **Time:** 30 seconds per question.

Trap	How to avoid it
Using the wrong digit position	Write “1st, 2nd, 3rd” above the digits.
Choosing the number with the highest last digit	Compare from the left.
Reversing the subtraction	“A is subtracted from B” is B − A.
Counting end symbols	The first and last items lack a neighbour.

Scope

Related PYQs / topics	Where
Series of numbers and letters with a pattern	Series (earlier topic)
Alphabet position questions	Alphabet Position and Word Manipulation (Topic 29)

CHAPTER 5

Practice Set

6 original questions; answers on the next page

A. Letter, number and symbol series

Q1. Refer to the series (left to right) and answer. T 6 M + 2 5 K A W Q 5 @ 4 D M @ How many symbols are immediately preceded by a letter and also immediately followed by a number?

A. None B. One C. Two D. Three

Q2. Refer to the series (left to right) and answer. 9 + # W K 8 # + & B M 6 3 * 7 A How many numbers are immediately followed by a symbol?

A. None B. One C. Two D. Three

Q3. Refer to the series (left to right) and answer. S R 9 # % 3 & 7 P $ N + 3 & P # How many letters have a symbol immediately before them and a symbol immediately after them?

A. Two B. None C. One D. Three

B. Three-digit number puzzles

Q4. This question is based on five three-digit numbers (left to right): 690 462 599 766 314. (Example: 697 has first digit 6, second digit 9, third digit 7.) What is the result if the third digit of the lowest number is subtracted from the third digit of the highest number?

A. 3 B. 1 C. 2 D. 4

Q5. This question is based on five three-digit numbers (left to right): 673 783 686 672 730. (Example: 697 has first digit 6, second digit 9, third digit 7.) What is the result if the first digit of the highest number is added to the second digit of the lowest number?

A. 15 B. 14 C. 13 D. 16

Q6. This question is based on five three-digit numbers (left to right): 696 734 397 727 750. (Example: 697 has first digit 6, second digit 9, third digit 7.) What is the product of the second digit of the highest number and the first digit of the lowest number?

A. 16 B. 14 C. 17 D. 15

Answer key

Q	Answer	Reason
1	B	Scanning the series from left to right and applying the condition gives one (checked by program).
2	D	Scanning the series from left to right and applying the condition gives three (checked by program).
3	A	Scanning the series from left to right and applying the condition gives two (checked by program).
4	C	Highest = 766 and lowest = 314. Result: 2 (checked by program).
5	B	Highest = 783 and lowest = 672. Result: 14 (checked by program).
6	D	Highest = 750 and lowest = 397. Result: 15 (checked by program).

CHAPTER 6

Quick Revision

One page for the last day

Item	Rule
Preceded by	Item on the left
Followed by	Item on the right
Digit positions	1st hundreds, 2nd tens, 3rd units
Highest and lowest	Compare from the hundreds digit
A subtracted from B	B − A

PYQ RULE: All 3 PYQs: read left to right; check both neighbours (Maths 2024 Q83); find the highest and lowest, then use the digit asked (Science 2024 Q81, Q94).

ROUTINE: Exam-day routine: read the condition; mark neighbours or find highest and lowest; compute. 30 seconds per question.